

- ① 오답.md 어제 오답부터 재풀이
- ② 신규는 공부계획 ID대로
- ③ 채점: solve-exam-prep 로 풀이 대조

떠올릴 개념

- Ch9·10·14 전 범위 (Ch6은 직병렬만)
- 미니모의 4 (9.94·10.42·14.23·14.40) 시간 재고
- 부호·rms/peak·½계수 최종 점검

문항 목차 (총 6문항)

- [1] 10.50 — 최대전력전달
- [2] 14.25 — 필터·공진
- [3] 9.94 — 정현파·phasor·임피던스 · 모의
- [4] 10.42 — 최대전력전달 · 모의
- [5] 14.23 — 필터·공진 · 모의
- [6] 14.40 — 필터·공진 · 모의

10.50 a) Assume that R_o in Fig. P10.49 can be varied between 0 and 10 k Ω . Repeat (b) and (c) of Problem 10.49.

PSPICE
MULTISIM

- b) Is the new average power calculated in (a) greater than that found in Problem 10.49(a)?
- c) Is the new average power calculated in (a) less than that found in 10.49(d)?

14.25 Using a 50 nF capacitor in the bandpass circuit shown in Fig. 14.22, design a filter with a quality factor of 5 and a center frequency of 20 krad/s.

DESIGN
PROBLEM

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- Specify the numerical values of R and L .
- Calculate the upper and lower cutoff frequencies in kilohertz.
- Calculate the bandwidth in hertz.

[생성 9.94 · 모의] 직렬 RLC — KVL 페이지 합 검증

$v(t) = 100 \cos(2\pi \cdot 60 \cdot t)$ [V] 가 $R = 20 \, \Omega$, $L = 50 \, \text{mH}$, $C = 20 \, \mu\text{F}$ 직렬 RLC 에 인가된다.

(a) X_L , X_C 를 구하라.

(b) $Z = R + j(X_L - X_C)$, $|Z|$, θ 를 구하고 유도성/용량성을 판정하라.

(c) 전류 I (rms) 를 구하라.

(d) V_R , V_L , V_C 를 구하고 $E = \sqrt{V_R^2 + (V_L - V_C)^2}$ 가 전원 전압과 일치하는지 검증하라.

[생성 문제 · 9~15주차 강의노트 범위 · 해답: 생성문제_해답.md]

— 풀이 —

10.42 The phasor voltage $\mathbf{V}_{ab}$ in the circuit shown in Fig. P10.42 is $300\angle 0^\circ$ V (rms) when no external load is connected to the terminals a, b. When a load having an impedance of $200 - j500 \Omega$ is connected across a, b, the value of $\mathbf{V}_{ab}$ is $156 - j42$ V (rms).

Problems **435**

- a) Find the impedance that should be connected across a, b for maximum average power transfer.
- b) Find the maximum average power transferred to the load of (a).
- c) Construct the impedance of part (a) using components from Appendix H if the source frequency is 500 Hz.

Figure P10.42

14.23 The input to the series RLC bandpass filter designed in Problem 14.21 is $5\cos\omega t$ V. Find the voltage drop across the series combination of the inductor and capacitor when (a) $\omega = \omega_o$; (b) $\omega = \omega_{c1}$; (c) $\omega = \omega_{c2}$; (d) $\omega = 0.1\omega_o$; (e) $\omega = 10\omega_o$.

14.40 The input to the RLC bandreject filter designed in Problem 14.39 is $10\cos\omega t$ V. Find the voltage drop across the series combination of the inductor and capacitor when (a) $\omega = \omega_o$; (b) $\omega = \omega_{c1}$; (c) $\omega = \omega_{c2}$; (d) $\omega = 0.125\omega_o$; (e) $\omega = 8\omega_o$.